Multiple Choice (10 marks)

1. Simplify and write the result with a rational denominator: $\sqrt{\sqrt[3]{\sqrt{\frac{1}{729}}}}$\$

(a) $\frac{}{\{3\sqrt{3}\}\{3\}}$

1. $\frac{1}{3}$

2. $\sqrt{3}$

3. $\frac{}{\{\sqrt{3}\}\{3\}}$

If there is a 70% chance that it will rain on a given day, what is the percent chance that it will NOT rain on that same day?

(a) 0%

(b) 30%

(c) 50%

(d) 70%

What is three fifth of 100?

(a) 3

(b) 5

(c) 20

(d) 60

The area bounded by the parabola y = x^2 and the lines y = 1 and y = 9 equals

(a) 8

(b) 84/3

(c) $64\sqrt{2}/3$

(d) 104/3

When n = 11, what is the value of 10 - (n + 6)?

(a) -7

(b) 5

(c) 7

(d) 27

True / False (10 marks)

Answer True or False

6. True or False: The relative maximum value of the function y = (ln x)/x is 1.
7. True or False: The quotient of 1,248 divided by 7 is 177 with a remainder of 9.
8. True or False: The point (6, 4) can lie on the circle that passes through the points (3, 4) and (5, 7).
9. True or False: Noah rides his bike at a speed of 10 miles per hour.
10. True or False: The P-value of the test for the hypothesis H 0: p=0.3 is 0.1446, indicating moderate evidence against H 0.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In an algebra test with 35 problems, a student must achieve at least an 80% score to pass. Discuss how many problems can be missed while still passing, and explain your reasoning.
12. An equilateral triangle has a side length of 8. Calculate the area of a square that shares the same perimeter as this triangle and explain your reasoning step by step.

End of Paper